

MATH 3312, SPRING 2026: MIDTERM 2

This exam has 6 (six) printed pages. Please show all your work. Outside materials are NOT permitted for use on this test. Do not spend more than 10 minutes on any one part of a problem. If you get stuck for more than a few minutes, just move on to the next part. *You should feel free to use the results of earlier problems in your solutions to later ones, regardless of whether you have finished them or not.* You have 50 minutes to work on the test. Write your answers in the space provided.

Name _____

	Score
Problem 1	/10
Problem 2	/10
Problem 3	/10
Problem 4	/10
Problem 5	/10
Total	/50

Name _____

(1) (10 points) **True or false. Write one or two lines justifying your answer:**

(a) There exists $A \in M_{5 \times 5}(k)$ whose characteristic polynomial has degree 6.

(b) There exists $A \in M_{3 \times 3}(k)$ such that $\min_A(x) = (x-1)^2$ and $\text{ch}_A(x) = x(x-1)^2$.

(c) There is a vector space V over a finite field k such that $|V| = 10$.

(d) Every degree n irreducible polynomial $f(x) \in \mathbb{F}_p[x]$ divides $x^{p^n} - x$.

(e) The columns of the matrix $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{bmatrix}$ form a basis for $\mathbb{Q}^3$.

Name _____

- (2) (10 points) Give examples of the following or explain why an example *cannot* exist:

(a) A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that is not upper triangulizable; that is, there is no basis $\mathcal{B}$ for $\mathbb{R}^2$ such that $[T]_{\mathcal{B}}$ is upper triangular.

(b) A matrix $A \in M_5(\mathbb{F}_{17})$ with minimal polynomial $x^5 + x^3 - x + 1$.

Name _____

- (3) (10 points) State the Rank-Nullity theorem and write down the key statements leading to its proof.

Name _____

- (4) (10 points) Find the Smith Normal Form of the matrix

$$A = \begin{bmatrix} 3 & 1+i \\ 1-i & 2 \end{bmatrix}$$

and use this to compute the invariant factors of $\mathbb{Z}[i]^2/\text{im } A$.

Name _____

- (5) (10 points) Give a formula for the number of conjugacy classes of 3×3 -matrices over a finite field k .